

31 — GENETIC ALGORITHM FOR 0/1 KNAPSACK PROBLEM

Aim

To solve the 0/1 Knapsack Problem using a Genetic Algorithm to maximise the total value of selected items while satisfying the weight constraint.

Software Required

- Python 3.x
- Jupyter Notebook / Google Colab
- Python Standard Library (random)

Procedure:

1. Define the weights, values, and capacity of the knapsack.
2. Represent each solution as a binary chromosome.
3. Generate an initial random population.
4. Calculate the weight and value of each chromosome.
5. Assign fitness based on the capacity constraint.
6. Select the fittest chromosomes.
7. Perform single-point crossover.
8. Apply mutation to the offspring.
9. Repeat the process for several generations.
10. Select and display the best feasible solution.

Code:

```
import random

weights = [2, 3, 4, 5]
values = [3, 4, 5, 6]
capacity = 8

def fitness(chromosome):
    weight = 0
    value = 0
    for i in range(len(chromosome)):
        if chromosome[i] == '1':
```

```

        weight += weights[i]
        value += values[i]
    if weight <= capacity:
        return value
    return 0

def ga_knapsack(pop_size=10, generations=30):
    population = [
        ''.join(random.choice('01') for _ in range(4))
        for _ in range(pop_size)
    ]
    for _ in range(generations):
        population = sorted(
            population,
            key=fitness,
            reverse=True
        )
        parents = population[:4]
        children = []
        while len(children) < pop_size:
            p1, p2 = random.sample(parents, 2)
            point = random.randint(1, 3)
            child = p1[:point] + p2[point:]
            child = ''.join(
                bit if random.random() > 0.1
                else str(1 - int(bit))
                for bit in child
            )
            children.append(child)
        population = children

```

```
best = max(population, key=fitness)

return best, fitness(best)

random.seed(1)

solution, value = ga_knapsack()

print("Best solution:", solution)

print("Value:", value)
```

Output:

```
-----, ----- ga_knapsack, -----

print("Best solution:", solution)
print("Value:", value)

... Best solution: 0101
Value: 10
```